Tabela surowych próbek dyskretnych – Optymalizacja dwutorowa 2D PSO Trend MF-BB

Zestawienie dyskretnych próbek trajektorii czasowych (t, v_C, i_L, s) dla regulatora Trend MF-BB z nastawami z optymalizacji 2D PSO (w_i* = 0.4351, fc_i* = 446.9 Hz, w_u* = 15.73, fc_u* = 454.3 Hz, T_s = 10 us).

Okno 1: Stan ustalony pod pełnym obciążeniem (240 V, 1.2 kW, t □ [145.0, 146.5] ms, krok 20 µs)

t [ms]	v_C [V]	i_L [A]	s		t [ms]	v_C [V]	i_L [A]	s
145.000	239.72	13.035	1	|	145.760	239.43	14.573	1
145.020	239.89	12.487	0	|	145.780	240.20	10.827	0
145.040	239.97	11.938	1	|	145.800	239.70	13.477	1
145.060	239.46	14.587	1	|	145.820	240.36	9.727	0
145.080	240.24	10.840	0	|	145.840	239.85	12.379	1
145.100	240.23	10.289	1	|	145.860	239.99	11.831	0
145.120	239.73	12.940	1	|	145.880	240.03	11.282	1
145.140	239.89	12.393	0	|	145.900	239.53	13.931	1
145.160	239.96	11.844	1	|	145.920	240.24	10.185	0
145.180	239.46	14.493	1	|	145.940	239.73	12.836	1
145.200	240.22	10.747	0	|	145.960	239.89	12.289	0
145.220	239.72	13.397	1	|	145.980	239.96	11.740	1
145.240	240.37	9.647	0	|	146.000	239.46	14.389	1
145.260	239.87	12.299	1	|	146.020	240.21	10.643	0
145.280	239.36	14.947	1	|	146.040	239.70	13.294	1
145.300	240.17	11.202	0	|	146.060	239.89	12.747	0
145.320	240.18	10.651	1	|	146.080	240.47	8.993	0
145.340	239.68	13.302	1	|	146.100	239.97	11.646	1
145.360	239.86	12.755	0	|	146.120	239.47	14.296	1
145.380	239.95	12.206	1	|	146.140	240.21	10.549	0
145.400	240.08	11.657	0	|	146.160	239.71	13.200	1
145.420	240.11	11.106	1	|	146.180	239.88	12.653	0
145.440	239.61	13.756	1	|	146.200	239.97	12.104	1
145.460	240.30	10.008	0	|	146.220	240.09	11.555	0
145.480	239.80	12.659	1	|	146.240	240.12	11.004	1
145.500	239.95	12.112	0	|	146.260	239.62	13.654	1
145.520	240.00	11.562	1	|	146.280	240.30	9.906	0
145.540	239.50	14.212	1	|	146.300	239.79	12.558	1
145.560	240.23	10.465	0	|	146.320	239.94	12.010	0
145.580	239.73	13.116	1	|	146.340	239.99	11.461	1
145.600	239.91	12.569	0	|	146.360	239.49	14.111	1
145.620	239.99	12.019	1	|	146.380	240.21	10.365	0
145.640	240.10	11.470	0	|	146.400	239.71	13.016	1
145.660	240.13	10.920	1	|	146.420	239.88	12.469	0
145.680	239.62	13.570	1	|	146.440	239.95	11.920	1
145.700	239.82	13.023	0	|	146.460	239.45	14.569	1
145.720	240.44	9.271	0	|	146.480	240.22	10.822	0
145.740	239.93	11.924	1	|	146.500	239.71	13.473	1

Okno 2: Stan przejściowy – skok zadanego napięcia 240 V → 160 V (t □ [149.8, 151.3] ms, krok 20 µs)

t [ms]	v_C [V]	i_L [A]	s	|	t [ms]	v_C [V]	i_L [A]	s
149.800	240.42	9.311	0	|	150.560	204.18	-20.649	0
149.820	239.92	11.963	1	|	150.580	202.69	-20.668	1
149.840	239.42	14.612	1	|	150.600	202.26	-17.975	1
149.860	240.19	10.867	0	|	150.620	199.91	-20.647	0
149.880	239.69	13.517	1	|	150.640	198.43	-20.609	1
149.900	240.35	9.767	0	|	150.660	198.01	-17.917	1
149.920	239.85	12.419	1	|	150.680	195.68	-20.475	0
149.940	239.99	11.871	0	|	150.700	194.22	-20.382	1
149.960	240.03	11.322	1	|	150.720	193.81	-17.689	1
149.980	239.53	13.972	1	|	150.740	191.52	-20.137	0
150.000	240.24	10.225	0	|	150.760	190.09	-19.988	1
150.020	240.57	6.469	0	|	150.780	188.73	-19.820	0
150.040	240.53	2.713	0	|	150.800	186.24	-22.127	0
150.060	240.12	-1.031	0	|	150.820	185.85	-19.432	1
150.080	239.33	-4.754	0	|	150.840	184.53	-19.209	0
150.100	238.17	-8.446	0	|	150.860	182.11	-21.405	0
150.120	236.65	-12.098	0	|	150.880	181.73	-18.712	1
150.140	234.77	-15.700	0	|	150.900	180.46	-18.435	0
150.160	232.53	-19.242	0	|	150.920	178.13	-20.525	0
150.180	229.95	-22.716	0	|	150.940	176.71	-20.197	1
150.200	229.47	-20.021	1	|	150.960	176.34	-17.505	1
150.220	228.99	-17.329	1	|	150.980	174.12	-19.488	0
150.240	226.61	-20.713	0	|	151.000	171.72	-21.407	0
150.260	225.06	-21.031	1	|	151.020	171.36	-18.713	1
150.280	224.59	-18.338	1	|	151.040	170.11	-18.298	0
150.300	222.13	-21.603	0	|	151.060	168.82	-17.868	1
150.320	220.55	-21.859	1	|	151.080	168.46	-15.179	1
150.340	220.08	-19.165	1	|	151.100	168.11	-12.494	1
150.360	218.69	-19.398	0	|	151.120	167.18	-12.046	0
150.380	216.15	-22.503	0	|	151.140	166.20	-11.587	1
150.400	215.69	-19.808	1	|	151.160	165.86	-8.907	1
150.420	214.28	-19.981	0	|	151.180	165.11	-8.435	0
150.440	211.69	-22.967	0	|	151.200	164.32	-7.954	1
150.460	211.24	-20.271	1	|	151.220	163.98	-5.278	1
150.480	210.80	-17.579	1	|	151.240	163.63	-2.606	1
150.500	208.46	-20.479	0	|	151.260	162.95	-4.290	0
150.520	206.97	-20.556	1	|	151.280	162.61	-1.619	1
150.540	206.53	-17.863	1	|	151.300	162.26	1.048	1

Okno 3: Stan ustalony po skoku pod obciążeniem (160 V, 538 W, t □ [170.0, 171.5] ms, krok 20 µs)

t [ms]	v_C [V]	i_L [A]	s	|	t [ms]	v_C [V]	i_L [A]	s
170.000	159.98	6.691	0	|	170.760	159.76	7.389	1
170.020	160.24	5.080	0	|	170.780	160.09	5.782	0
170.040	160.33	3.466	0	|	170.800	160.25	4.170	0
170.060	159.99	6.126	1	|	170.820	160.10	4.695	1
170.080	160.01	6.651	0	|	170.840	159.77	7.354	1
170.100	160.26	5.040	0	|	170.860	160.09	5.746	0
170.120	160.35	3.425	0	|	170.880	160.25	4.135	0
170.140	160.02	6.086	1	|	170.900	160.10	4.660	1
170.160	160.03	6.611	0	|	170.920	159.76	7.318	1
170.180	160.28	4.999	0	|	170.940	160.08	5.712	0
170.200	160.36	3.384	0	|	170.960	160.24	4.100	0
170.220	160.03	6.045	1	|	170.980	159.90	6.760	1
170.240	160.04	6.570	0	|	171.000	160.16	5.151	0
170.260	160.28	4.957	0	|	171.020	160.26	3.539	0
170.280	160.36	3.343	0	|	171.040	159.93	6.199	1
170.300	160.02	6.003	1	|	171.060	159.95	6.725	0
170.320	160.04	6.528	0	|	171.080	160.21	5.115	0
170.340	160.27	4.916	0	|	171.100	160.30	3.502	0
170.360	160.35	3.302	0	|	171.120	159.97	6.162	1
170.380	160.01	5.963	1	|	171.140	159.99	6.688	0
170.400	160.02	6.488	0	|	171.160	160.24	5.076	0
170.420	160.26	4.876	0	|	171.180	160.33	3.463	0
170.440	160.33	3.262	0	|	171.200	160.00	6.123	1
170.460	159.99	5.923	1	|	171.220	160.01	6.648	0
170.480	160.00	6.449	0	|	171.240	160.26	5.036	0
170.500	160.23	4.838	0	|	171.260	160.35	3.422	0
170.520	160.30	3.225	0	|	171.280	160.02	6.083	1
170.540	159.96	5.885	1	|	171.300	160.03	6.608	0
170.560	159.97	6.411	0	|	171.320	160.28	4.995	0
170.580	160.20	4.801	0	|	171.340	160.36	3.381	0
170.600	160.08	5.326	1	|	171.360	160.03	6.041	1
170.620	160.06	5.851	0	|	171.380	160.04	6.566	0
170.640	160.23	4.240	0	|	171.400	160.28	4.954	0
170.660	160.09	4.765	1	|	171.420	160.36	3.340	0
170.680	159.75	7.424	1	|	171.440	160.02	6.000	1
170.700	160.08	5.817	0	|	171.460	160.04	6.525	0
170.720	160.24	4.206	0	|	171.480	160.27	4.913	0
170.740	160.10	4.730	1	|	171.500	160.35	3.299	0